

Saving the Oysters



A Plan to Solve the Oyster Crisis in the Chesapeake Bay

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Introduction

The Chesapeake Bay is arguably the most important body of water in the United States. It is a significant contributor not only to the economy and ecosystem of Maryland, but to the economy and ecosystem of the entire nation. Many millions of people live around the Chesapeake Bay, with major cities such as Baltimore and Annapolis relying upon its quality daily. Additionally, there are several thousands of jobs which are dependent upon it. The health and maintenance of the Chesapeake Bay is of utmost importance.

Should its quality diminish, the state of Maryland and the United States in turn would experience a plethora of treacherous consequences.

To the right is a map of the Chesapeake Bay watershed. It can be seen that the Bay has a direct effect on the waterways of six different states: Maryland, Virginia, West Virginia, Delaware, Pennsylvania, and New York. The Bay is thus intrinsic to the ecosystems of all of these states, which make up a significant portion of the entire mid-Atlantic United States.

Several of these consequences have already been taking place over the past decades. Specifically, the oyster population of the Chesapeake Bay has reduced steadily and significantly since the 17th century due to disease, pollution, and overharvesting. The Statistics Department at the University of Maryland, Baltimore County, recognizes this issue, and would like to issue a formal proposal to the Chesapeake Bay

Foundation for an investigation which will greatly contribute to the final solving of the oyster problem. Our plan is to collect data from various locations within the Chesapeake Bay and its watershed, measuring several variables such as salinity, turbidity, and dissolved oxygen. This data will be used extensively in linear regression models, which will provide a greater understanding of the primary causes of the reduced oyster population. That is, we will isolate the most significant variables, facilitating developments to improve the conditions, and ultimately nullifying them. We have confidence that our process will be effective in saving the oyster population of the bay and in its overall health.



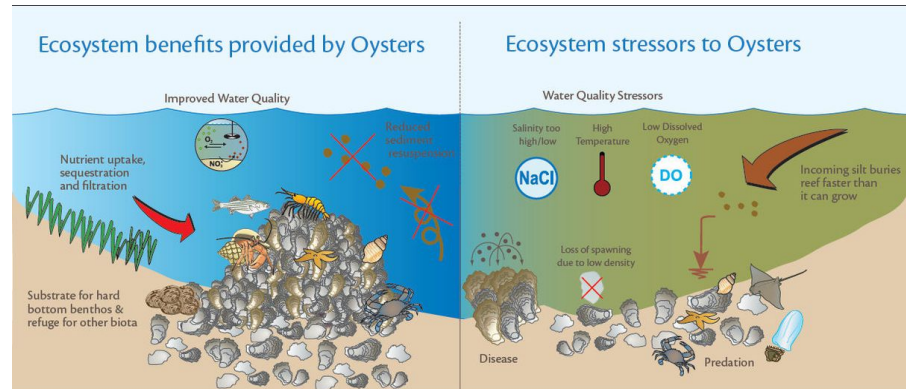
Statement of Need

A number of factors have contributed to the reduced oyster population in the Chesapeake Bay, mainly as a result of disease, overharvesting, and most importantly, pollution. Much of the pollution of the bay comes from the runoff of toxins from factories and households alike. This runoff has significant adverse effects on the organisms living in the bay, and thereby the entire watershed. Pollutants, if they do not directly kill the organisms, can make regions uninhabitable, resulting in *dead zones*. These zones are defined as “areas of water bodies where aquatic life cannot survive because of low oxygen levels.” (Environmental Protection Agency). Dead zones

would become ubiquitous if the runoff of pollutants continued, dramatically deteriorating the health of the Chesapeake Bay.

Runoff directly affects the bay's oyster population by reducing the number of habitable zones they can occupy. This is significant because of their status as a *keystone species*. A keystone species is a species of organism which is definitive for an entire ecosystem. That is, if an ecosystem lost its keystone species, the ecosystem would be "dramatically different or cease to exist altogether."

(National Geographic). Oysters are largely responsible for the filtration of the Chesapeake Bay, that is they "remove light-blocking algae from Bay waters, facilitate removal of excess nutrients, and improve water quality for other organisms." (Chesapeake Bay Foundation). Thus, oysters are a critical component of the Chesapeake Bay ecosystem, providing healthy habitats for fish, crabs, and many other species. The importance of oysters to fish and crab populations indicates their significance in the seafood industry, an area the state of Maryland has heavily relied upon for centuries.



According to the National Oceanic and Atmospheric Administration (NOAA), only 1-2% of the native oyster population remains in the Chesapeake Bay, a figure that is certain to sink further should significant action not be taken against it. The Chesapeake Bay Foundation (CBF) has done incredible work in combating the problem, already having dedicated major portions of its budget toward oyster restoration. The oyster population has increased since these efforts have begun, but several challenges prevent the CBF from fully resolving the issue.

In order to prevent any further deterioration of the Bay's habitats and to contribute to the full restoration of its health, the precise and most pertinent issues must be determined before deciding upon future courses of action. Should this not be done, it is more likely that the oyster population will continue to struggle, diminishing the quality of the Chesapeake Bay as a whole. As mentioned before, an unhealthy Chesapeake Bay would have many direct detrimental effects on at least five states, and indirect detrimental effects on the United States.

Goals and Objects

1. Restore oyster populations to ideal levels
 - a. Determine the most pertinent variables in diminishing oyster habitats.
 - b. Isolate which pollutants are significantly affecting oyster populations.
 - c. Ascertain the primary source of pollutants along with its subsidiaries.
2. Maintain health of thousands of species living in the Chesapeake Bay

- a. Provide suitable habitats for organisms by restoring the natural filtration conducted by oysters.
 - b. Eliminate dead zones by nullifying runoff of toxins and pollutants.
 - c. Clean portions of the bay most directly affected by pollution. Impose stricter regulations on fishing and harvesting to support the health of local habitats.
3. Generate increased awareness of the significance of the oyster problem
 - a. Publish data and analyses so that information is easily accessible.
 - b. Broadcast findings on local newscasts, displaying statistics and showcasing current developments.
 - c. Explain the importance of the issue through advertisements and the use of engaging visuals which are easily understood.
4. Provide job opportunities to thousands of workers in the Mid-Atlantic region
 - a. Pay stipends to volunteers who assist in gathering of data and in habitat clean-ups.
 - b. Improve the seafood industry by augmenting the overall health of the Chesapeake Bay, helping restaurants to generate more revenue and have more success.
 - c. Demonstrate the value of contributing to restoration efforts, thereby inspiring people to become more involved in supporting local ecosystems and thus the economy as a whole.

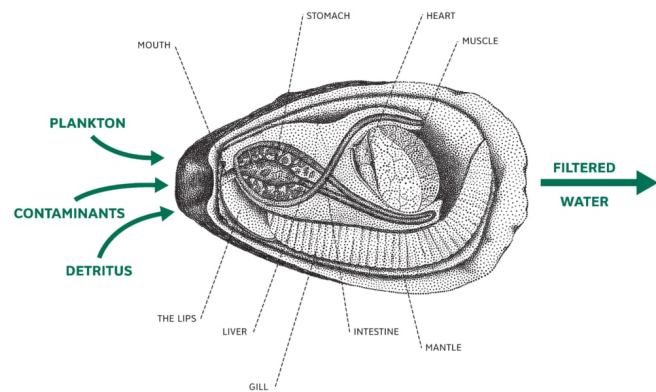
Literature Review

Due to the Chesapeake Bay's significance, there have been many studies on its water quality and the effect it has on wildlife. All studies reaffirm the importance of oysters to the bay. The Chesapeake Bay Foundation itself has made oyster restoration a top priority for several years. The Foundation cites oysters as a critical component of the bay ecosystem. They write of oysters' role in providing habitats for crab and fish species, and in "removing light-blocking algae from Bay waters" (CBF), which helps to remove extra nutrients such as nitrogen and phosphorus. However, the oyster population has reduced dramatically compared to where it was originally. As a result, many efforts have been made in the hopes of restoring the population. Success in doing so would allow the Chesapeake Bay to thrive once again by vastly improving the quality of its waters and wildlife, in addition to augmenting local economies. While there are many factors which may be considered to achieve this overall goal, it is certain that resolving the oyster problem will be a significant step towards it.

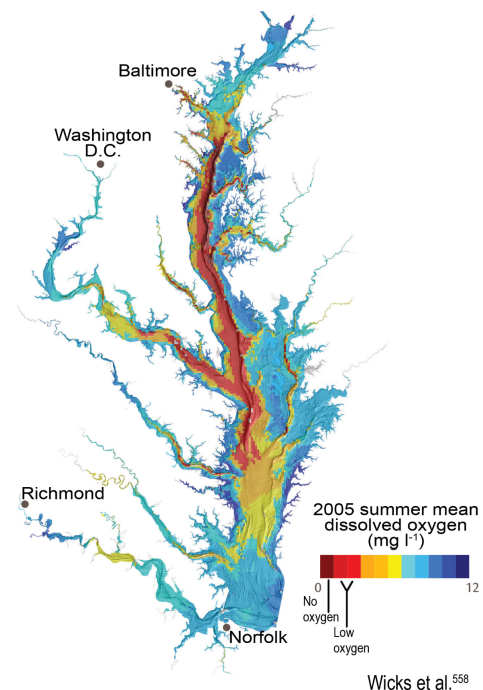


As mentioned before, oysters are designated as a keystone species, or a species which is extremely important and integral to an ecosystem. If such a species were to be removed or drastically altered, the ecosystem would suffer greatly. In extreme cases, the ecosystem may seem to exist entirely. Challenges to the oyster population in the Chesapeake Bay include but are not limited to: pollution, overharvesting, and disease. It was stated previously that these factors have led to only 1-2% of the native oyster population remaining (NOAA). Below is a simple diagram depicting the filtration system which oysters provide. Oysters are classified as filter

feeders, meaning they feed on harmful materials such as contaminants and *detritus* from water, and expel cleaner, filtered water as a result. Detritus is defined as waste or debris, in the case of oysters, it generally refers to the remains or excrements of other organisms living within the bay (Britannica) . As stated, oysters help to remove toxins and other detrimental factors from the water in the Bay, thereby improving the overall quality of the water. If there are more oysters living in the Chesapeake Bay, then the water will be filtered much more quickly and efficiently, allowing wildlife to thrive by providing suitable habitats with healthy water for other organisms.



There are various factors which directly affect oyster populations, but the most prominent would be pollution. The majority of this pollution comes from the runoff of toxins from factories and households. This occurs when rainwater picks up contaminants such as pet waste, fertilizers, pesticides, oil, etc. The rainwater, carrying the contaminants, makes its way into sewage systems, then into rivers, and ultimately feeds into the Chesapeake Bay. While much of the contaminants becomes diluted, that is the water possesses a decreasing concentration of pollutants at each step, their effects can still be dramatic enough to significantly harm the bay. According to the CBF, runoff has three primary detrimental effects on the Chesapeake Bay. The first is that the contaminants carried in the water can reshape the watershed by “destabilizing the natural contours of streams” and “altering their depths” (CBF). The second is perhaps the most trivial, in that the pollutants affect the quality of water by contaminating it, which in turn makes treatment of the water far more expensive. Thus, runoff can have direct implications on the economy by complicating the process of producing drinking water. The third reason refers to the harm caused to aquatic life, wherein “eroded dirt...blocks sunlight from reaching underwater grasses” (CBF) and covers the habitats of oysters and other organisms. Runoff has several other adverse effects on the Chesapeake Bay environment, all of which possess devastating potential to the entire watershed.



Another factor which affects oyster populations are water indices such as salinity, dissolved oxygen, pH, and others. Salinity is measured as the amount of dissolved salt present in a body of water. It is known

Climate change is likely to expand and intensify “dead zones,” areas where bottom water is depleted of dissolved oxygen because of nitrogen pollution, threatening living things.

that oysters require a relatively low level of salinity in order to thrive, typically about 14-28 ppt (Museum of the City of New York). This is a small range, so maintaining the salinity of the bay has proven to be difficult. Further, it should be noted that the Chesapeake Bay inherently has a higher salinity than many bodies of water due to its size and proximity to the ocean. Salinity tends to decrease the further the body of water is from the ocean, but since the Bay feeds directly into the open ocean, its salinity typically ranges from 25-30 ppt (Chesapeake Bay Program). Next is *dissolved oxygen*, which is the oxygen that is present in water and available for organisms (MCNY). As the vast majority of living organisms require oxygen to some extent, its availability in water is crucial. Oysters can survive lower levels of dissolved oxygen than most other organisms, but still require at least 20% saturation. Much of the Chesapeake Bay has relatively low dissolved oxygen levels, though, because of its great depth in many locations. The picture on the previous page is a map of the Chesapeake Bay with colors representing the amount of dissolved oxygen in each area. Oyster populations are also affected by pH levels, which measure the acidity of water. Specifically, oysters rely upon carbonate to build their shells. If water becomes too acidic, their shells would become thinner and make them more susceptible to predators, ultimately decreasing their population (NOAA).

Additional factors include overharvesting and disease, both of which have further reduced oyster populations in the Chesapeake Bay. In the mid-Atlantic region and in Maryland particularly, oysters are a major part of food culture often enjoyed as a delicacy. As a result, they are often harvested at rates which far exceed what would be considered environmentally responsible. According to a study by the Florida museum, overharvesting has been occurring for many centuries, but has worsened in recent years with the use of modern technologies. While these technologies are more efficient, they often cause significant physical damage to ecosystems, and do not take into account oysters' ability to "sustain a healthy population size" (Florida Museum). Disease is yet another contributor to worsened oyster conditions. Namely, the Virginia Institute of Marine Science cites *dermo* and *MSX* as major threats to the oyster population. *Dermo* is a parasite which thrives in high-temperature waters, and rapidly multiplies, spreads, and kills oysters. *MSX* is another parasite which is dependent upon specific salinities and temperatures. In waters of higher salinity, *MSX* is particularly effective at attacking oysters, making it a major concern in the Chesapeake Bay. Controlling both overharvesting and the spread of disease are necessary in restoring oyster populations.



While there have been many efforts in oyster restoration in the 21st century, the factors discussed still pervade the Chesapeake Bay and routinely affect oysters. However, a 2020 report from Maryland's Department of Natural Resources showed that the number of large oysters had increased from 300 million in 2018 to 400 million in 2020. The Chesapeake Bay Foundation remains concerned, though, because the increase in population may be temporary. The Foundation writes that the estimated number of young oysters (less than one year old) is the sixth lowest since 1999 (CBF). Additionally, the NOAA has provided funding for organizations to conduct recovery efforts in several tributaries. Specifically, it has partnered with the Chesapeake

Bay Program in the hopes of restoring oysters in 10 tributaries by 2025. Despite the fact there are ongoing efforts to recover the oyster population to its once-healthy values, it is important to eliminate any sense of complacency. That is, the effort must continue to improve and increase in scope, which provides us with the best change to complete such a daunting task.

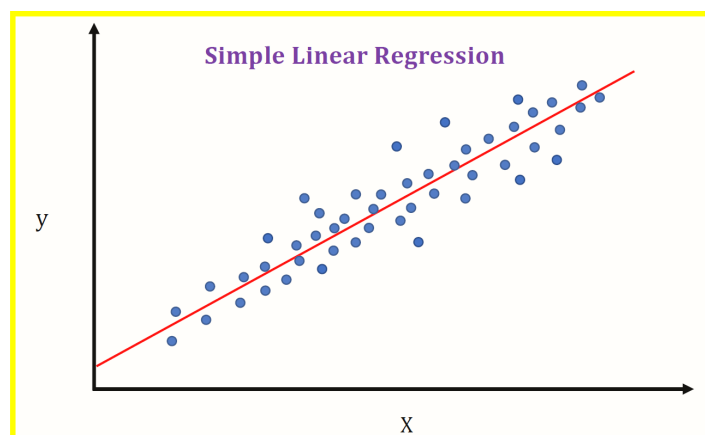
Project Description

Our plan is to explore the effects of several variables such as salinity, dissolved oxygen, turbidity, and others on the water quality of the Chesapeake Bay. By analyzing the variables, we will determine which is most impactful upon the oyster populations which are entirely dependent on relatively high water quality. Variable analysis will be completed through simple linear regression models.

We plan on spending two weeks recruiting qualified volunteers to assist in data collection. The recruitment process will consist of television and internet advertisements, radio announcements, and newspaper articles. The project could additionally be spread by word of mouth. We will have 50 volunteers who will be briefed on the purpose and importance of the project in addition to learning how to conduct accurate and precise measurements.

Firstly, we will spend one month collecting data in 100 predetermined locations within the Chesapeake Bay watershed. We will have ten separate crews on ten separate boats, each crew consisting of five volunteers, a driver, and two supervisors. Each of these crews will conduct measurements and record their findings and observations at ten separate locations. Measurements will consist of: oyster population within the designated circle around the area, salinity, pH, turbidity, and dissolved oxygen. We will also measure the concentrations of various pollutants in the areas. It is important to have a large sample size for locations because it will make the analysis more reliable due to the limitation of results occurring from random chance. The wide variety of locations will also give the data a great deal of divisibility. That is, we will be able to ascertain the effects of variables on the bay as a whole and on specific types of areas.

Next, we will begin linear regression analyses of the data in order to determine which of the variables are most significant. All of the aforementioned measurements will be included as variables in the regression, and by using simple techniques the significance of each will be compared. If one variable is deemed more significant than another, it may be logical for more resources to be allocated to it. For example, if we find salinity to be more significant in affecting oyster population than turbidity, future restoration efforts will seek to control salinity rather than turbidity. While the effects of many of the mentioned variables are already known, our investigation will reinforce



preexisting knowledge. Additionally, the investigation may assist organizations in planning future projects in the most effective way by avoiding the over-allocation to any particular area.

The final step would be to assess the effectiveness of the project. This will be done by repeating the data collection step in the precise way it was completed originally, i.e. collecting measurements of the same variables in the same locations. This step will be completed two years after the initial data collection, which provides enough time for recovery efforts to continue and for significant changes to be observed. The new data for each specific location can then be compared with its original data, giving us an understanding of how the situation has developed over time. Further, the differences in measurements will be subjected to an additional linear regression analysis which will tell us whether the new findings are statistically meaningful.

Budgets, Costs, Funds Requested

	Cost
Boats	\$750,000
Measuring Devices	\$35,000
Volunteer Stipend (\$100 for each day worked)	\$100,000
Advertisements	\$100,000
Publishing of Findings	\$15,000
Total	\$1,000,000

Note: Boats and measuring devices can potentially be provided by the CBF itself, significantly reducing costs.

Materials Needed

	Quantity
Volunteers	50
Boats	10
Salinometers	50
Dissolved Oxygen Meter	50
Turbidity Instruments	50

Project Assessment

The project proposed in this document will be assessed in a number of ways, not all of them trivial. Considering the primary goal is to contribute to oyster restoration in the Chesapeake Bay, the first point of assessment will be in comparing data from the beginning of the project to data at the end of the project. If the data shows that oyster populations have increased in the time

between the beginning and end of the project, it could be concluded that the project was effective. However, this assessment will be strengthened by a rigorous statistical analysis of the results. That is, the UMBC Statistical Department will determine whether the increase or decrease in population was statistically significant. If there is any observed decrease in oyster population, the project will be deemed unsatisfactory, especially if said decrease is shown to be statistically significant. Conversely, if the findings suggest a statistically significant increase in oyster population, it will be deemed successful.

While difficult to determine the precise timeline for the project due to several unpredictable variables, its assessment will take place two years after the analyses have been completed. This will provide enough time for the findings to be applied to restoration efforts and for the oyster population to change in a potentially meaningful way. Additionally, this time frame will limit the probability that the results were due to random chance, increasing its usefulness and reliability. Further, the findings and project assessments will be provided to the public regardless of the outcome, because awareness of the problem is one of the most important aspects in resolving it. Lastly, lessons learned from the project will certainly be applicable to future restoration efforts. The Chesapeake Bay Foundation can use our project assessment in order to improve its own plans by determining areas for improvement.

Timeline

May 20, 2022 - May 29, 2022

- Advertisements shown, raising awareness
- Recruitment of volunteers for clean-ups and data collection

May 30, 2022 - June 24, 2022

- Collecting data
 - Training volunteers how to use measuring devices
 - Recording precise, accurate, and detailed measurements

June 25, 2022 - August 22, 2022

- Clean up of most affected areas
 - Discarding trash such as plastics, cellophane, metals from habitats
 - Removing rust and general restoration of sewage systems
- Analysis of data using linear regression
 - Isolating most impactful variables
 - Determining which pollutants are most detrimental to oyster health

August 23, 2022 - May 29, 2024

- Continued bay clean-up
- Restoration efforts continued
- Findings from analysis applied to plans combating reduced oyster population
 - Results published locally and nationally

May 30, 2024 - June 24, 2022

- Collecting new data in precise locations from previous study
 - Recording precise, accurate, and detailed measurements
 - Comparing observations and findings to those made previously

June 25, 2024 - July 9, 2024

- Analysis of new data using linear regression
 - Compare results with original study
 - Assess usefulness of project
- Publish new findings and assessment of project

July 10, 2024 - Future

- Ongoing clean-up and maintenance of the Chesapeake Bay
- Procedures and recovery efforts based on results of project

Project Summary

The UMBC Statistics Department is optimistic that the project proposed in this document will be effective in aiding the restoration of the oyster population in the Chesapeake Bay. As such, it seeks funds from the Chesapeake Bay Foundation in order to collect data and complete analyses. The bay has and always will be an entity of incredible importance to the state of Maryland and to the United States as a whole. The project will not only increase public awareness of this importance, but it will also be successful in helping to maintain the overall health of the Bay by supporting oyster restoration efforts. Ideally, the project will contribute to the augmentation of the job market and of organism health as well.

Although two years is a relatively short period of time to see significant results, we are confident that our studies will provide an improved understanding of the causes of the diminishing oyster population. This improved understanding will in turn allow several organizations to make plans regarding the future of the bay, not only with respect to oyster restoration but also with respect to the thousands of different species that live within it.

Our goal is that in the very near future we will be able to look back upon this project and apply it once again in the hopes of recovering the oyster population permanently. That is, the results and methods of this project will be repeatable in the future, potentially becoming a reliable source of ecosystem recovery efforts around the world. While our investigation seeks to resolve a specific problem in a specific location, its methods are applicable to a wide variety of environmental issues, such as air pollution, deforestation, and others. However, it cannot be applied to these situations until it is adequately assessed. That is, the project will only be successful if the health of the bay is improved and maintained far into the future.

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